

4. SOME POWER COMPARISONS

We compare the power of our FDR controlling procedure with some other Bonferroni-type procedures which control the FWER. Under the overall null hypothesis the proposed method controls the FWER at level q^* . We take the FWER controlling methods and the FDR controlling method to control the FWER weakly at the same level, using $q^* = \alpha$, and compare the power of the methods from the two different approaches under different configurations. It is clear from the comment in Section 2.1, property (b), that a method which controls the FDR is generally more powerful than its counterpart which controls the FWER (in the strong sense). The magnitude of the difference remains to be investigated.

4.1. *The Setting*

We studied this question by using a large simulation study, where the family of hypotheses is the expectations of m independent normally distributed random variables being equal to 0. Each individual hypothesis is tested by a z -test, and the test statistics are independent. We use $q^* = \alpha = 0.05$. The configurations of the hypotheses involve $m = 4, 8, 16, 32$ and 64 hypotheses, and the number of truly null hypotheses being $3m/4, m/2, m/4$ and 0 . The non-zero expectations were divided into four groups and placed at $L/4, L/2, 3L/4$ and L in the following three ways:

- (a) linearly decreasing (D) number of hypotheses away from 0 in each group;
- (b) equal (E) number of hypotheses in each group;
- (c) linearly increasing (I) number of hypotheses away from 0 in each group.

These expectations were fixed (per configuration) throughout the experiment.

The variance of all variables was set to 1, and L was chosen at two levels—5 and 10—thereby varying the signal-to-noise ratio.

Each simulation involved 20000 repetitions. The estimated standard errors of the power are about 0.0008–0.0016. As the same normal deviates were used in a single repetition across all configurations with the same number of hypotheses, and the alternatives in different configurations were monotonically related, a positive correlation was induced. This correlation reduces the variance of a comparison between two methods or two configurations to below twice the variance of a single method.

4.2. *Results*

Fig. 1 presents the estimates of the average power (the proportion of the false hypotheses which are correctly rejected) for three methods. The two FWER controlling methods, the Bonferroni (dotted curves) and Hochberg's (1988) method (broken curves), are compared with the new FDR controlling procedure (full curves). The following observations can be made from the results displayed.

- (a) The power of all the methods decreases when the number of hypotheses tested increases—this is the cost of multiplicity control.
- (b) The power is smallest for the D-configuration, where the non-null hypotheses are closer to the null, and is largest for I (which is obvious but mentioned for completeness).

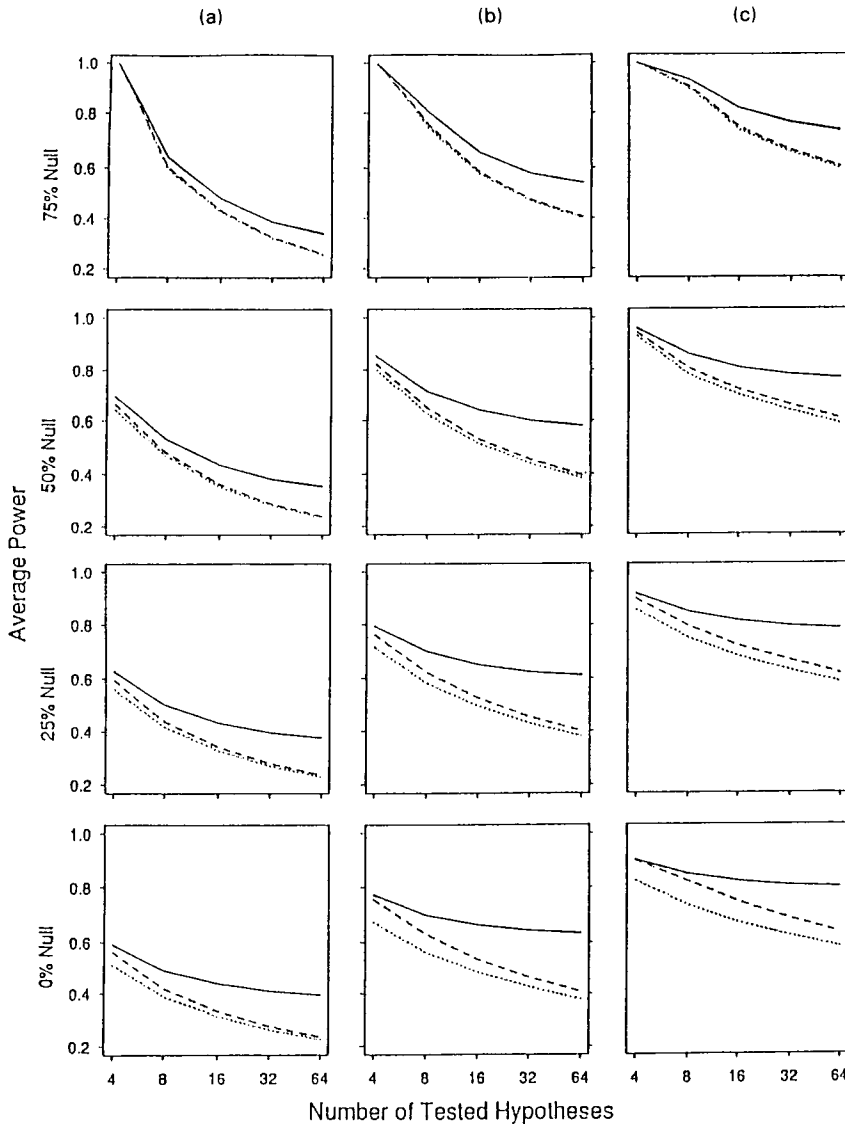


Fig. 1. Simulation-based estimates of the average power (the proportion of the false null hypotheses which are correctly rejected) for two FWER controlling methods, the Bonferroni (·····) and Hochberg's (1988) (----) methods, and the FDR controlling procedure (—): (a) decreasing; (b) equally spread; (c) increasing

- (c) The power of the FDR controlling method is uniformly larger than that of the other methods.
- (d) The advantage increases with the number of non-null hypotheses.
- (e) The advantage increases in m . Therefore, the loss of power as m increases is relatively small for the FDR controlling method in the E- and I-configurations.

- (f) The advantage in some situations is extremely large. For example, testing 32 hypotheses, equally spread in four clusters from 1.25 to 5 so that none is true, the power of the Bonferroni method is 0.42. The procedure suggested increases the power to 0.65; testing as few as four hypotheses, half of which are true, the values are 0.62 and 0.70 respectively.
- (g) It is known that Hochberg's method offers a more powerful alternative to the traditional Bonferroni method. Nevertheless, it is important to note that the gain in power due to the control of the FDR rather than the FWER is much larger than the gain of the FWER controlling method over the Bonferroni method.

Casting these results into a different form, it may be seen that in some configurations up to half the non-null hypotheses which were not rejected by the Bonferroni procedure are now rejected by the FDR controlling method, when at least half of the tested hypotheses are non-null. Even when only 25% of the hypotheses are non-null, the gain in power is such that about a quarter of the equally spaced hypotheses which were not rejected before are now rejected.

Fig. 1 allows us also to answer a question raised by a referee, about how $E(V/m_0)$ is controlled by the FDR controlling method. This error rate is 0 when $m_0 = 0$ and otherwise can be approximated by the average level α_{ave} at which the individual hypotheses are tested. Obviously α_{ave} is always less than q^* , but for m_0 away from m more can be said: let R_{ave} be the average number of rejections and f_{ave} be the average power (displayed in Fig. 1). It follows that $m\alpha_{\text{ave}} \leq R_{\text{ave}}q^* \equiv (m_0\alpha_{\text{ave}} + m_1f_{\text{ave}})q^*$, so

$$\alpha_{\text{ave}} \leq f_{\text{ave}}q^* \frac{m_1}{m_1 + m_0(1 - q^*)}.$$

Therefore $E(V/m_0)$ looks as in Fig. 1 but is smaller by a factor of q^* or even less. For $m = m_0$ the error is much closer to q^*/m_0 than to q^* : 0.0132, 0.0063, 0.0033, 0.0017 and 0.0009 for the four, eight, 16, 32 and 64 hypotheses tested respectively.

5. CONCLUSION

The approach to multiple significance testing in this paper is philosophically different from the classical approaches. The classical approach requires the control of the FWER in the strong sense, a conservative type I error rate control against any configuration of the hypotheses tested. The new approach calls for the control of the FDR instead, and thereby also the control of the FWER in the weak sense. In many applications this is the desirable control against errors originating from multiplicity.

Within the framework suggested, other procedures may be developed, including procedures which utilize the structure of specific problems such as pairwise comparisons in analysis of variance. A different direction, which we have already pursued, is to develop an adaptive method which incorporates the ideas of Hochberg and Benjamini (1990). In this paper, however, we have only focused on presenting and motivating the new approach that calls for controlling the FDR, and we have demonstrated that it can be developed into a simple and powerful procedure. Thus the cost paid for the control of multiplicity need not be large. This might contribute

considerably to the proliferation of a greater awareness of multiple-comparison problems, and of cases where something is done about it.

ACKNOWLEDGEMENTS

We would like to express our thanks to J. P. Shaffer and J. W. Tukey, whose comments and questions about an earlier draft helped us to crystallize the approach presented here. We thank Yetty Varon for her participation in the programming of the simulation study and Yechezkel Kling for pointing out a need to tighten the original proof of theorem 1.

APPENDIX A: PROOF OF LEMMA

The proof of the lemma is by induction on m . As the case $m = 1$ is immediate, we proceed by assuming that the lemma is true for any $m' \leq m$, and showing it to hold for $m + 1$.

If $m_0 = 0$, all null hypotheses are false, $\mathbf{Q}$ is identically 0 and

$$E(\mathbf{Q} | P_1 = p_1, \dots, P_m = p_m) = 0 \leq \frac{m_0}{m+1} q^*.$$

If $m_0 > 0$, denote by P'_i , $i = 1, 2, \dots, m_0$, the p -values corresponding to the true null hypotheses, and the largest of these by $P'_{(m_0)}$. These are $U(0, 1)$ independent random variables. For ease of notation assume that the m_1 p -values that the false null hypotheses take are ordered $p_1 \leq p_2 \leq \dots \leq p_{m_1}$. Finally, define j_0 to be the largest $0 \leq j \leq m_1$ satisfying

$$p_j \leq \frac{m_0 + j}{m+1} q^*, \quad (4)$$

and denote the right-hand side of inequality (4) at j_0 by p'' .

Conditioning on $P'_{(m_0)} = p$,

$$\begin{aligned} E(\mathbf{Q} | P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) &= \int_0^{p''} E(\mathbf{Q} | P'_{(m_0)} = p, P_{m_0+1} = p_1, \dots, \\ &\quad P_m = p_{m_1}) f'_{P'_{(m_0)}}(p) dp \\ &\quad + \int_{p''}^1 E(\mathbf{Q} | P'_{(m_0)} = p, P_{m_0+1} = p_1, \dots, \\ &\quad P_m = p_{m_1}) f'_{P'_{(m_0)}}(p) dp \end{aligned} \quad (5)$$

with $f_{P'_{(m_0)}}(p) = m_0 p^{(m_0-1)}$.

In the first part $p \leq p''$. Thus all $m_0 + j_0$ hypotheses are rejected, and $\mathbf{Q} \equiv m_0/(m_0 + j_0)$. Evaluating the integral first, and then using inequality (4), we obtain

$$\frac{m_0}{m_0 + j_0} (p'')^{m_0} \leq \frac{m_0}{m_0 + j_0} \frac{m_0 + j_0}{m+1} q^* (p'')^{m_0-1} = \frac{m_0}{m+1} q^* (p'')^{m_0-1}. \quad (6)$$

In the second part of equation (5), consider separately each $p_{j_0} < p_j \leq P'_{(m_0)} = p < p_{j+1}$, along with $p_{j_0} \leq p'' < P'_{(m_0)} = p < p_{j_0+1}$. It is important to note that, because of the way by which j_0 and p'' are defined, no hypothesis can be rejected as a result of the values of p , p_{j+1} , p_{j+2} , $\dots$, p_{m_1} . Therefore, when all hypotheses—true and false—are considered together, and their p -values thus ordered, a hypothesis $H_{(i)}$ can be rejected only if there exists k , $i \leq k \leq m_0 + j - 1$, for which $p_{(k)} \leq \{k/(m+1)\} q^*$, or equivalently